

Time: 2½ Hours

Total Marks: 75

Q.1 Attempt ANY THREE from the following: (15M)

- (a) With the help of Key components of Business analytics, explain Business Analytics.
- (b) Explain challenges and Application of business analytics.
- (c) Explain Technologies and Tools in Business Analytics.
- (d) Write a short note on Measure of central tendency and Measure of Variability.
- (e) Explain relationship among variable in terms of Correlation analysis and regression analysis in applied business analytics.
- (f) Explain types of distribution with example.

Q.2 Attempt ANY THREE from the following: (15M)

- (a) Explain in brief business intelligence tools used for Data Analysis.
- (b) Explain probability and probability distribution.
- (c) Explain Application of Probability and Probability distribution in Business analytics.
- (d) What is Decision-making under uncertainty? Explain types of uncertainty in Business Decision-making.
- (e) Write a short note on decision making model under uncertainty in Applied Business analytics.
Explain various elements of decision analysis.
- (f) Write the application of Decision making under uncertainty in Business Analytics.

Q.3 Attempt ANY THREE from the following: (15M)

- (a) Define sampling and explain its importance in business analytics, highlighting its role in making inferences about populations based on sample data.
- (b) Why sampling is important in Business analytics?
- (c) What is Hypothesis testing? Discuss the types of Hypothesis test in Business analytics.
- (d) Why sampling Distribution is important in business analytics? Explain.
- (e) What is confidence interval estimation? Explain concept of confidence interval estimation.
- (f) Discuss the steps to construct a confidence interval in business analytics.

Q.4 Attempt ANY THREE from the following:

(15M)

- (a) What is regression analysis? Explain the different types of regression analysis.
- (b) Discuss the application of regression analysis in Business analytics.
- (c) Discuss the benefits and challenges in regression analysis in brief.
- (d) Explain the term estimating the relationship in business analytics and its application.
- (e) Explain regression analysis in terms of statistical inference.
- (f) Explain time series analysis and forecasting in business analytics.

Q.5 Attempt ANY THREE from the following:

(15M)

- (a) Define optimization modelling and its key element in business analytics.
- (b) Discuss the types of optimization problem and application of optimization modelling in business analytics
- (c) What is Analysis of Variance (ANOVA)? Also discuss the important concept in ANOVA.
- (d) What is Simulation modelling? Discuss the key aspect of simulation modelling.
- (e) Explain Experimental design and its role in business analytics
- (f) Explain different types of Experimental Designs in Business analytics.

4. Attempt any three of the following:

15

- a Explain the concept of Confusion Matrix and accuracy of classifier. Why accuracy alone may not always be a reliable performance metric?
- b Describe the ROC (Receiver Operating Characteristic) curve and AUC (Area Under the Curve).
- c How is Cross-Validation applied in Machine Learning models? Explain with examples.
- d Define the concept of Clustering and explain at least three real-world applications of Clustering.
- e Discuss the types of Hierarchical Clustering methods in detail.
- f Explain the various linkage methods used in Hierarchical Clustering.

5. Attempt any three of the following:

15

- a. What is K-Means Clustering? Explain its working with an example.
- b. Write a detailed note on Dimensionality Reduction techniques.
- c. Explain the term Principal Component Analysis (PCA). How does it help in reducing dimensionality?
- d. Provide a brief overview of Market Basket Analysis and its significance.
- e. Explain the main steps involved in Apriori Algorithm with a suitable example.
- f. Discuss the strengths and weaknesses of K-Means Clustering. In which situations is it most effective?

Q 4) Attempt any three from the following:

[15]

- a) What are the concerns regarding performance and battery life while writing code for embedded system?
- b) Explain different types of libraries for embedded Systems which work with limited memory.
- c) Explain the terms Business Model Canvas?
- d) What is venture capital? How can one exit?
- e) Describe Hobby Projects and open source?
- f) Describe Funding an Internet of things startup?

Q 5) Attempt any three from the following:

[15]

- a) Explain the steps for Manufacturing PCBs.
- b) What is Designing Kits?
- c) Discuss different environmental issues in Internet of Things
- d) Explain five critical requirements for sensor common projects?
- e) Explain Charactering the Internet of Things?
- f) Write a short note on Mass-Producing the case and other Fixtures.

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N. B.:(1) All questions are compulsory.

(2) Make suitable assumptions wherever necessary and state the assumptions made.

(3) Answers to the same question must be written together

(4) Numbers to the right indicate marks.

(5) Draw neat labeled diagrams wherever necessary.

(6) Use of Non-programmable calculators is allowed.

Q 1) Attempt any three from the following:

[15]

- Define and explain the Internet of Things with 2 examples.
- Explain Calm and ambient technology using example of Live wire?
- List and explain the Roles of the people making IOT?
- What are TCP and UDP Ports? Explain with example?
- Discuss the issue of privacy in Internet of things?
- Explain the following application layer protocols: HTTP, HTTPS, SMTP, FTP, POP3, IMAP

Q 2) Attempt any three from the following:

[15]

- Explain microcontrollers and system-on-chips with respect to embedded computing.
- Explain the difficulties encountered during the transition from a prototype to mass production?
- Difference between Open sources versus Closed Source?
- Explain Beagle bone Black with a focus on the following aspects: Operating System, Programming Language, Debugging, Hardware and Openness.
- Compare python characteristics with C++.
- Explain Open source has a competitive advantage and disadvantages .Discuss.

Q 3) Attempt any three from the following:

[15]

- Define and explain Laser Cutting? Explain the main features to consider while choosing a laser cutter.
- What CNC Milling? Explain
- Classify and explain Types of 3D Printing?
- What is Comet? Discuss.
- Define Clockodillo?
- Explain the following terms with respect to APIs: (a) Scraping, (b) Mashing up APIs.

[Total Marks: 75]

(6) Use of **Non-programmable** calculators is **allowed**.

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5. Attempt Any three of the following:

- a. What is Congestive Heart Failure? Explain its different symptoms.
 - b. What are the common symptoms of breast cancer? How can early detection improve outcomes?
 - c. Explain the different ML applications for Cancer.
 - d. Explain the relationship between healthcare analysis and internet.
 - e. What is Deep Learning? Explain the role of deep learning in healthcare analysis.
 - f. What kind of ethical issues comes in the healthcare in deep learning?
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(Time: 2½ Hours)

[Total Marks: 75]

N. B.: (1) All questions are compulsory.(2) Make suitable assumptions wherever necessary and state the assumptions made.(3) Answers to the same question must be written together.(4) Numbers to the right indicate marks.(5) Draw neat labeled diagrams wherever necessary.(6) Use of Non-programmable calculators is allowed.**1. Attempt Any three of the following:**

15

- What is healthcare analytic? Explain its different applications.
- Explain healthcare acts in healthcare industry in briefly.
- How visualizations help healthcare professionals understand and communicate complex patient data?
- Explain the different role of healthcare policies in healthcare industries.
- Write short note on Electronic Medical Record-1.
- Why social history of patient is important in healthcare analysis?

2. Attempt Any three of the following:

15

- What is the SOAP? How does it help in documenting patient care?
- What is ICD? Explain the features of ICD.
- Explain the method of probabilistic reasoning in healthcare analysis.
- What is Machine Learning? Explain the use of machine learning pipeline in healthcare analysis.
- Write a short note on NDC.
- Explain the term "predicting mortality for cardiology practice" in healthcare analytics.

3. Attempt Any three of the following:

15

- How Hospital Value-Based Purchasing program uses to measure the quality of healthcare?
- Explain the use of Hospital Readmission Reduction program in healthcare analysis.
- Write a short note on ESRD in healthcare data analysis.
- Explain the use of Merit Based Incentive Payment System (MIPS).
- Explain the role of HAC in the condition of a patient was not present or incubating at the time of admission.
- Write a short note on Healthcare Effectiveness Data and Information Set (HEDIS).

4. Attempt Any three of the following:

15

- Explain the concept of Predictive analytics in healthcare.
- Why splitting data into train and test set is important in machine learning?
- What is Blood Pressure? Explain its different types.
- How machine learning help on predicting the discharge status of patients in the Emergency Department (ED)?
- Write a short note on "miscellaneous information in healthcare".
- Explain the Neural Network in machine learning in healthcare.